Session 4: Synthesis, Q&A, and Pathway to Continued Learning

From Techniques to Operational EO AI in the Philippines

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Session Plan

- **Duration:** 2 hours (120 minutes)
- **Plan:**
  - 0–15: 4-day journey recap
  - 15–45: Technique selection matrix
  - 45–75: Best practices for deployment
  - 75–100: Philippine EO platforms & continued learning
  - 100–115: Action plans (3-month roadmap)
  - 115–120: Closing & feedback

Session Objectives

- Connect the techniques from Days 1–4 into one decision framework
- Apply deployment best practices: data quality, validation, monitoring
- Know where to find Philippine EO data and further learning
- Leave with a concrete 3-month action plan

What You Can Now Do

- Build LSTMs for EO time series (Day 4)
- Train RF, CNNs, U-Net for EO tasks (Days 2–3)
- Use foundation models, embeddings, SSL and XAI to scale and explain models (Day 4)
- Design operational pipelines and monitoring (Day 4)

Recap of Days 1–4

Highlights

- Day 1: Copernicus and Philippine EO data, AI/ML basics, Python geospatial, Google Earth Engine
- Day 2: Random Forest land cover (Palawan), CNNs, transfer learning
- Day 3: U-Net segmentation, Sentinel-1 flood mapping, object detection
- Day 4: LSTM drought forecasting (Mindanao), foundation models, embeddings, SSL, XAI

Technique Selection Matrix

- Classification (image/patch) → CNN / transfer learning, or a classifier on embeddings
- Segmentation (pixel-wise) → U-Net / GeoFM fine-tuning
- Detection (bounding boxes) → YOLO/SSD/DETR
- Time series → LSTM/Transformers, temporal validation
- Tabular + EO features → Random Forest/XGBoost

Best Practices

Data-Centric AI

- Fix labels, balance classes, represent regions/seasons

Validation

- Temporal splits, spatial CV, stratified sampling

Monitoring

- Concept/data drift, retraining schedule, dashboards

Deployment

- APIs, performance budgets, documentation, XAI

Platforms & Continued Learning

Philippine EO Platforms

- **CopPhil Infrastructure** (Copernicus mirror site hosted at PhilSA): Sentinel data, JupyterLab, infra.copphil.philsa.gov.ph
- **CopPhil Digital Campus**: self-paced EO courses, courses.copphil.philsa.gov.ph
- **SkAI-Pinas / DOST-ASTI:** DIMER (model repository), AIPI
- PhilSA Space+ Data Dashboard, NAMRIA Geoportal, PAGASA climate data

Keep Learning

- All course materials stay online: anefelo.github.io/eo-ai-ml-training
- Earth Engine tutorials, EO College, TorchGeo / TerraTorch docs
- Learn with colleagues: pick a shared project, review each other’s notebooks

3-Month Action Plan

- Week 1: Pick a starter project, run baseline
- Month 1: Proof of concept with validation
- Month 2: Improve data + model, stakeholder review
- Month 3: Pilot deployment + monitoring

Closing & Feedback

- What will you try first? Share one idea with a neighbour
- Feedback and questions: dimitris@anefelo.com
- Thank you!